

Jack Misliniski

Software engineer · Founder, Aevox: AI analytics deployed in 5 labs · NYU CS · MSc Applied Physiology · ex-Barclays
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SUMMARY

Engineer who builds and ships AI software end to end, and the operator who deploys it. I built Aevox, a metabolic-testing analysis platform, and run every deployment myself: the first sales call, ingesting each lab's raw exports as they actually arrive, shipping fixes the same week, and folding them back into one core. Five labs live on five cart vendors, from a CommonSpirit Health / EXOS performance center to independent labs. My edge: as an exercise physiologist, I can read the test the software analyzes. NYU Computer Science; three years building analytics and trading credit at Barclays.

EXPERIENCE

Aevox — Founder & engineer

2024 – Present

- **Architecture.** React/TypeScript app with the analysis engine running in the browser and a thin serverless API for auth, storage, sharing and model calls (Vercel, Clerk, Neon Postgres). 900+ tests; the engine went from v2 to v25 since July, and saved reports are stamped with the engine version that produced them.
- **Ingest with AI under guardrails.** Reads a lab's export as uploaded: unit-aware mapping onto 31 fields and a time scale measured from the data, not trusted from labels. When the rules refuse a sheet, Claude proposes a column map that is accepted only if the full columns pass physics checks: human ranges, time that advances, heart rate that tracks VO_2 , and the cart's own identities ($RER = VCO_2/VO_2$, constant body mass, $VE = RR \times VT$). Reads COSMED, MGC, PNOE, VO_2 Master and KORR exports.
- **Deterministic engine, model as reviewer.** Ventilatory thresholds come from an ensemble of detectors; the model's read is bounded, and nothing renders until a physiologist confirms. Each of 19 report cards is measured, provisional or withheld, with the reason on page one.
- **Evals and regression.** An LLM reviewer reads each new report, and recurring findings become engine rules. Each engine bump records a before/after diff across 32 real test sessions.
- **Deployment.** One isolated, branded instance per lab (own project, auth and database), provisioned by one script, plus a keep-local mode that keeps clinical data in the browser. Trial users' asks ship in hours: a lab's hand-drawn lactate comparison became a product card the same week; a cart export the engine had never read was supported the day the lab sent it.
- **Go-to-market and operations.** Outbound to labs (100+ cold emails, 22% reply rate), discovery, pricing and contracts. I run the company on scheduled Claude agents I built behind one ops API; agents draft and watch, and nothing reaches a customer without my sign-off.

Barclays Investment Bank — Associate, Investment Grade Credit Trading

New York, NY · 2022 – 2025

- Built and owned the desk's analytics stack (Python, SQL, kdb/q) for PnL attribution, volumes, market share and risk. Market-made and managed risk on a **\$1B book** of front-end IG corporate bonds with real-time coverage of 20+ buy-side accounts. **Tradeweb 2023 Top 5 Performer.**

PROJECTS AND RESEARCH

Parsel — Healthcare × AI hackathon, winning entry

2026

A voice digital biomarker for Parkinson's: a self-referenced baseline and drift, never a diagnosis. Built the clinician dashboard and patient app (React, TypeScript, ElevenLabs voice) on a FastAPI + Baseten backend. Live at parsel.health.

Mitochondrial oxidative capacity outpaces redox buffering — manuscript in preparation

2026

Multi-omics analysis of the MoTrPAC endurance-training dataset in Python: 85 redox genes, 42,770 observations across 19 tissues, 5 omic layers and 4 time points. Pipeline public at github.com/jackmis610/motrpac-redox.

bioML — metabolic prediction from wearable signals

2025

Synthetic-data generator plus a multitask LightGBM model predicting VO_2 , RER and substrate oxidation from PPG heart rate, power and CGM. On a synthetic cohort (subject-wise 5-fold CV, $n=100$): RER MAE 0.034, LT2 AUC 0.953.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, kdb/q.

AI / LLM: Claude API, tool use, structured outputs, prompt caching, physics-checked model proposals, LLM review and evals, agent orchestration with Claude Code and scheduled agents.

Systems: React, Vite, Vercel serverless, Clerk, Postgres (Neon), Cloudflare Workers, WebSockets, vitest, GitHub Actions.

ML / data: pandas, scikit-learn, LightGBM, PyTorch; time-series models on noisy biosignals, grouped cross-validation. **Physiology:** CPET, ventilatory and lactate thresholds, substrate oxidation, instrument validation.

EDUCATION AND ADDITIONAL

M.S. Applied Physiology, University of Colorado Colorado Springs · GPA 4.0 · 2025 – 2027. Mitochondrial bioenergetics and metabolic-cart validation (6.5% VO_2 bias vs. reference); two manuscripts in preparation.

B.A. Computer Science, New York University · 2020 – 2022. **United States Military Academy, West Point**, CS major · 2018 – 2020: Dean's List every semester, Cadet First Sergeant (top 2 of ~1,100).

Ultramarathoner (Tahoe Rim Trail 100, Run Rabbit Run 50) and Pacific Crest Trail thru-hiker (2,650 mi). Spanish and Portuguese.